

Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	FinRelief - AI
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Secure user signup and login using JWT-based authentication and password hashing.</i>	HIGH
2	Authorization levels	<i>Authenticated users can securely access and manage only their own financial and loan data.</i>	HIGH
3	External interfaces	<i>Integrates Gemini AI API for negotiation strategies, settlement guidance, and letter generation.</i>	HIGH

4	Transactions processing	<i>Processes loan, income, debt, and settlement data through validated backend APIs and stores records in SQLite.</i>	HIGH
5	Reporting	<i>Provides financial health insights, debt stress analysis, loan priority, settlement estimates, and negotiation outputs</i>	HIGH
6	Business rules, etc.	<i>Calculates financial ratios, prioritizes loans, estimates settlements, and applies rule-based fallback logic when needed.</i>	HIGH
7	Compliance to laws or regulations	<i>Protects user financial data through secure authentication, password hashing, input validation,</i>	HIGH
8	<i>Other</i>	<i>Negotiation strategies and settlement letters based on the borrowers Generates financial profile.</i>	HIGH

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Dashboard, financial calculations, and standard API operations should respond quickly.</i>	<i>Standard pages load within 3 seconds; normal API responses within 2 seconds.</i>
2	Scalability	<i>Modular React and FastAPI architecture should support future growth in users, loans, and AI features.</i>	<i>New modules can be added without major restructuring.</i>
3	Security & Data Privacy	<i>Protect passwords, authentication tokens, financial records, and API keys using secure mechanisms.</i>	<i>Hashed passwords, JWT authentication, protected routes, and .env-based secret management.</i>
4	Reliability & Availability	<i>System should handle invalid input and AI service failures without crashing.</i>	<i>Input validation, exception handling, and rule-based fallback available.</i>
5	Usability & Accessibility	<i>Provide a responsive, clear, and borrower-friendly interface across common devices.</i>	<i>Core workflows usable on desktop and mobile with clear navigation and validation feedback.</i>
6	Other	<i>Maintain modular, readable, and deployment-ready frontend and backend code.</i>	<i>Separate modules for services, utilities, APIs, and UI components; production build completes successfully.</i>